Network Topology and Finite-Horizon Diffusion Dynamics

A Reproducible Computational Study of Random, Scale-Free, and Small-World Networks

Gordon Zou

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Abstract

Network topology can change how quickly information spreads, but comparisons require a specified transmission mechanism, time horizon, and outcome measure. This study examines synchronous susceptible-infected (SI) diffusion on Erdos-Renyi (ER), Barabasi-Albert (BA), and Watts-Strogatz (WS) networks. Each network contains 800 nodes, with mean degree approximately eight. We use 5 independent graphs per configuration, 30 diffusion trials per graph and probability, and 21 sampled transmission probabilities, producing 22,050 trial outcomes across seven configurations. At a 10-round horizon, the fitted 50% adoption midpoints are 0.139 for ER and 0.111 for BA, while both have a sampled 80% adoption threshold of 0.20. WS configurations with little rewiring fail to reach 80% adoption within the horizon, even at certain transmission. The regular lattice's coverage at probability one is explained exactly by its finite-radius neighborhood. We report pointwise graph-level uncertainty, per-trial seeds, and reproducible analysis artifacts. The findings describe finite-horizon dynamics under a stylized mechanism; they do not establish a causal clustering effect, a phase transition, or predictions of real-world misinformation behavior.

1 Introduction

Misinformation motivates questions about how network structure enables rapid spread. Synthetic networks can help isolate assumptions and build experiments, but they do not encode content truth, platform incentives, or individual beliefs. The scientific question addressed here is narrower: **under one transmission process, how does network topology change mean adoption within a fixed horizon?**

Random-graph, preferential-attachment, and small-world models provide contrasting structures [1-3]. Prior cascade and social-contagion research motivates the importance of structure [4,5], but those studies use different mechanisms and do not validate the SI model used here.

The contribution is a transparent experimental workflow: a single simulation implementation shared by the command-line runner and notebook, approximately degree-matched network comparisons, independent graph replication, explicit outcome definitions, and auditable numerical records. The report is an exploratory simulation study, not a peer-reviewed publication or empirical analysis of platform data.

2 Methods

2.1 Network construction

All reference networks contain $n = 800$ nodes. ER networks use $G(n, 8/(n-1))$, giving expected mean degree eight. BA networks use preferential attachment with four edges added for each arriving node, giving mean degree close to eight. WS networks start from a ring lattice with eight neighbors per node and rewiring probabilities $\beta \in \{0, 0.01, 0.1, 0.5, 1\}$.

The comparison approximately matches average degree; it does not match degree sequences, path lengths, clustering, or realized edge counts. Disconnected networks are retained. Graph-level mean degree, clustering, and largest-component coverage are recorded rather than assumed equal. WS rewiring preserves the edge count but changes multiple structural properties simultaneously.

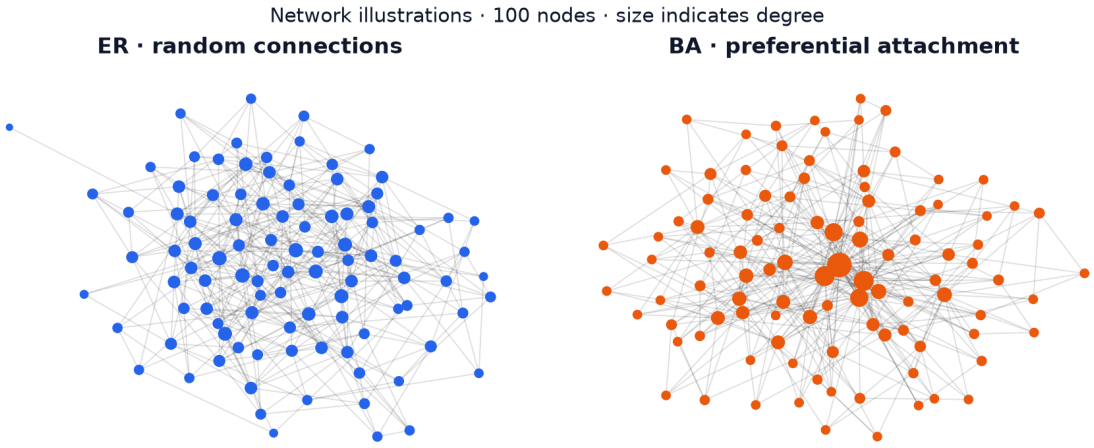


Figure 1: Seeded 100-node illustrations of ER and BA networks. Node size indicates degree. These illustrations are smaller than the experimental networks.

2.2 Transmission process

Let I_t be the adopted set after round t. At time zero, one uniformly sampled node is adopted. Each node in I_t independently attempts transmission to each neighbor outside I_t with probability p. Successful transmissions join I_{t+1} simultaneously. Adoption is permanent, and previously adopted nodes can attempt transmission again in later rounds.

For a susceptible node with k adopted neighbors, the conditional probability of adoption in one round is

$$\Pr(v \in I_{t+1} \mid v \notin I_t, k) = 1 - (1-p)^k.$$

The outcome is the adopted fraction at the configured horizon T,

$$A_T(p) = \frac{|I_T|}{n}.$$

We set $T = 10$. A quiet round does not terminate the process because subsequent repeated attempts can succeed. Newly adopted nodes cannot transmit until the next round. With $p = 0$, the

outcome is $1/n$; with $p = 1$, adoption reaches the source's graph-distance-T neighborhood. This repeated-attempt SI process differs from an independent-cascade model with a single transmission opportunity per newly active edge.

2.3 Replication and uncertainty

For every configuration we sample 5 independent graphs and perform 30 trials per graph at each of 21 probabilities on an equally spaced grid from zero to one. Each trial samples a source and a fresh transmission trajectory. The master seed is 42; all graph and diffusion seeds are retained. Runs at different probabilities are independently seeded rather than paired by source or transmission draws.

Let $Y_{g,r}(p)$ denote adoption for graph g and trial r. We first calculate the within-graph mean, $\bar{Y}_g(p)$, then the mean across graphs, $\bar{A}_T(p)$. If $s_G(p)$ is the sample standard deviation of graph-level means, the reported pointwise interval is

$$\bar{A}_T(p) \;\pm\; t_{0.975,G-1}\frac{s_G(p)}{\sqrt{G}}.$$

Intervals are clipped to $[0, 1]$. They describe uncertainty in mean adoption across graph sampling and finite within-graph simulation, not individual cascade ranges. With only 5 graph replicates, coverage relies on a coarse Student-t approximation. The intervals are pointwise, not simultaneous bands. We do not perform a formal significance test of topology differences.

2.4 Two different outcome summaries

The empirical threshold is

$$\hat{p}_{80} = \min\{p \text{ on the tested grid} : \bar{A}_T(p) \geq 0.8\}.$$

It is undefined when the criterion is never reached. This sampled statistic has resolution 0.05 and is reported without a threshold confidence interval. It is a finite-horizon operational measure, not a critical point established by finite-size scaling.

For curves that bracket 50% adoption, we fit the descriptive logistic function

$$f(p; \alpha, p_0) = \frac{1}{1 + \exp[-\alpha(p - p_0)]}$$

by nonlinear least squares, with $0 \leq \alpha \leq 1000$ and $0 \leq p_0 \leq 1$. The parameter p_0 is the fitted 50% midpoint, not $\hat{p}_{80}$. These are unweighted fits to mean adoption values, not logistic regression on individual binary observations. We record mean squared error and omit unsupported fits for flat curves or curves that do not bracket 50% adoption. Fit uncertainty is not estimated.

3 Results

3.1 ER and BA comparison

Table 1 reports the reference summaries. The realized mean degrees are 7.928 for ER and 7.960 for BA. ER's mean largest-component fraction is 0.99975; BA networks are connected in the sampled reference set.

Network	Mean degree	Clustering	Fitted midpoint	Sampled 80% threshold
ER	7.928	0.010	0.139	0.20
BA	7.960	0.041	0.111	0.20
WS ($\beta = 0$)	8.000	0.643	-	NR
WS ($\beta = 0.01$)	8.000	0.626	-	NR
WS ($\beta = 0.1$)	8.000	0.473	0.242	0.35
WS ($\beta = 0.5$)	8.000	0.087	0.147	0.20
WS ($\beta = 1$)	8.000	0.009	0.141	0.20

Table 1: Descriptive summaries of the reference experiment. NR means the 80% criterion was not reached on the sampled grid. A dash denotes an unreported midpoint because the curve did not bracket 50% adoption. Clustering is averaged across graphs; it is not a controlled causal treatment.

BA has an earlier fitted midpoint than ER in this run. At $p = 0.10$, mean adoption is 42.5% for BA and 20.0% for ER. At $p = 0.15$, the corresponding values are 78.7% and 59.9%. At $p = 0.20$, both exceed 80% mean adoption. Their sampled thresholds therefore coincide despite different midpoints.

Probability	ER mean [95% interval]	BA mean [95% interval]
0.10	20.0 [19.3, 20.8]	42.5 [36.5, 48.5]
0.15	59.9 [55.5, 64.2]	78.7 [73.3, 84.1]
0.20	89.8 [88.1, 91.5]	96.2 [94.3, 98.1]

Table 2: Mean adopted fraction and pointwise 95% graph-level intervals at selected probabilities. Values are percentages of nodes. These intervals do not represent uncertainty in the fitted midpoint or threshold.

BA hubs and graph-distance structure are plausible contributors to earlier adoption under this mechanism. The experiment does not isolate their individual contributions. A finer probability grid may distinguish threshold behavior that the current grid cannot resolve.

3.2 Watts-Strogatz rewiring

The regular lattice and the $\beta = 0.01$ setting never reach 80% mean adoption within 10 rounds on the tested grid. At $p = 1$, their mean adoption fractions are 0.10125 and 0.42005, respectively. For $\beta = 0.1$, the sampled threshold is 0.35; for $\beta = 0.5$ and $\beta = 1$, it is 0.20.

For the regular lattice, each node connects to its four nearest neighbors in each direction. Certain transmission advances at most four ring positions in either direction per round. Since $8T + 1 < n$ for the reference parameters, the number reachable after ten rounds is exactly

$$|I_{10}| = 2 \times 4 \times 10 + 1 = 81, \qquad A_{10}(1) = 81/800 = 0.10125.$$

This matches the recorded outcome. Failure to reach 80% in this configuration is therefore a direct consequence of the finite horizon and graph distances, not evidence that eventual diffusion is impossible on a connected lattice.

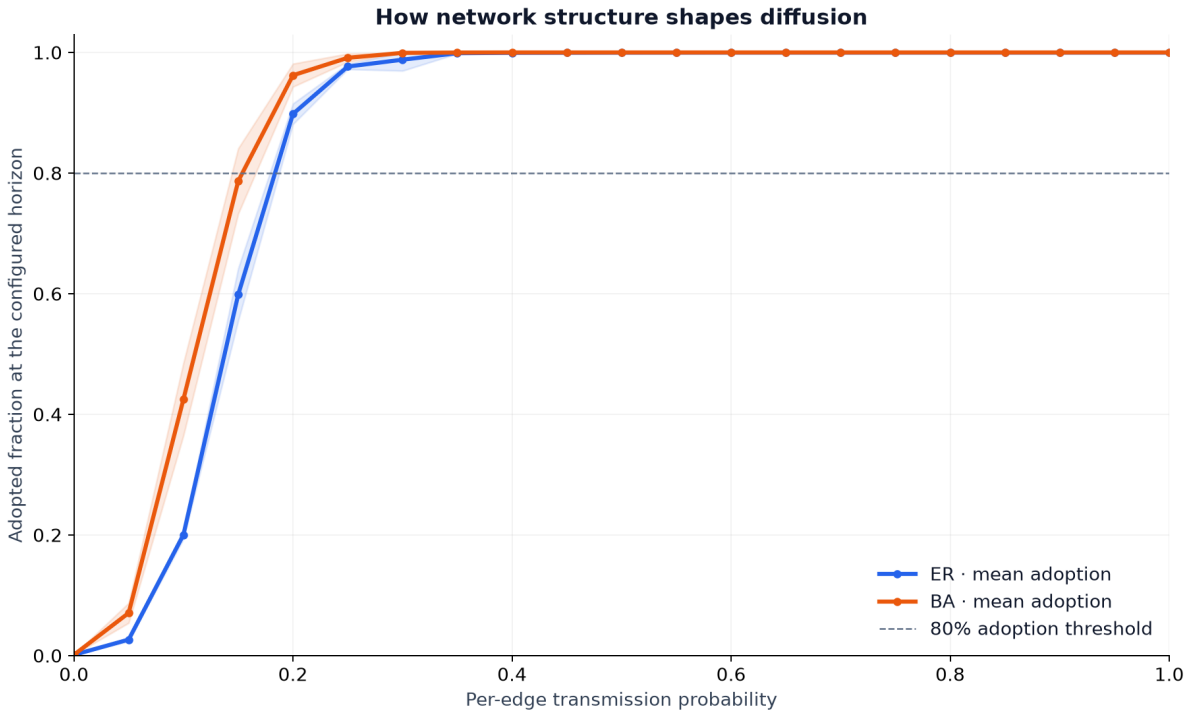


Figure 2: Reference ER and BA adoption curves. Shading represents pointwise 95% Student-t intervals across graph-replicate means. The horizontal line marks the 80% criterion.

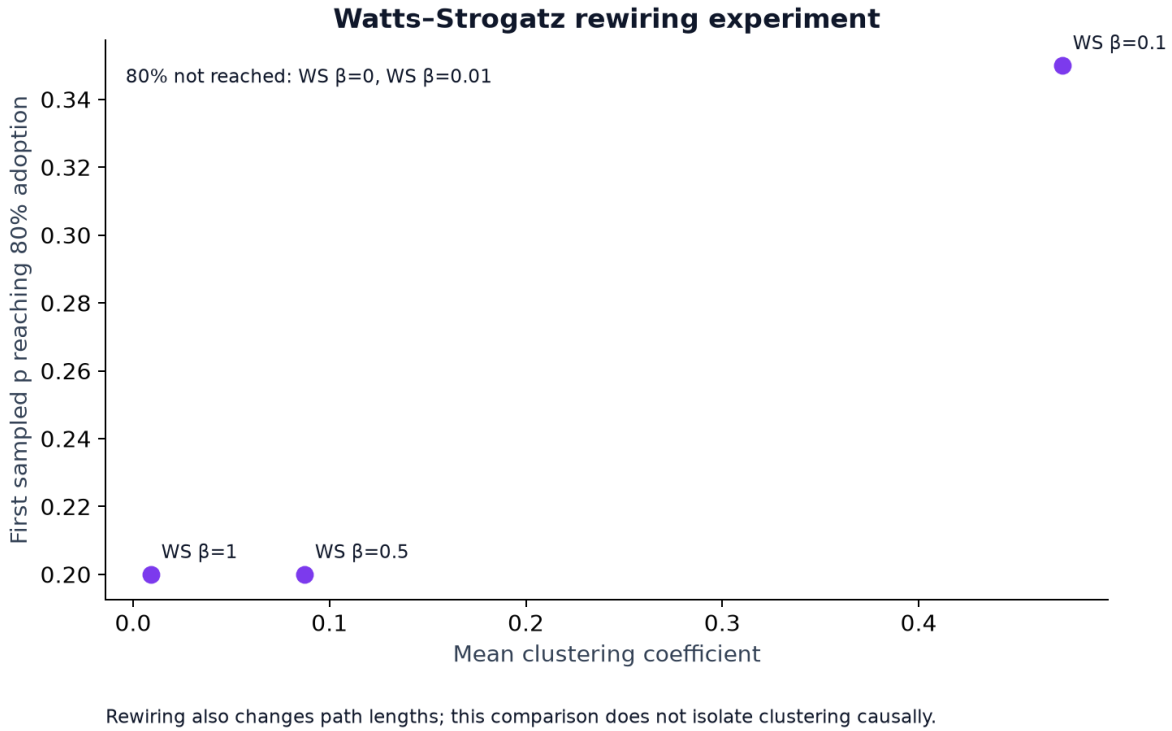


Figure 3: WS rewiring comparison. Configurations that do not reach 80% adoption are listed explicitly. Rewiring changes both clustering and other structural properties.

Lower clustering accompanies lower sampled thresholds across the shown WS settings, but rewiring also alters distances and shortcut structure. These observations cannot identify a causal effect of clustering alone. The ER/BA comparison likewise does not hold all other network properties fixed.

3.3 Fit interpretation

The fitted slopes are 36.05 for ER and 36.38 for BA, with descriptive mean squared errors 0.000015 and 0.000091. Small errors indicate agreement with these sampled mean curves; they do not establish mechanistic validity, external predictive accuracy, or the existence of a phase transition. No midpoint is reported for the two low-rewiring WS configurations because their curves never bracket 50% adoption.

4 Discussion

The experiment shows that the same local transmission process produces different finite-horizon adoption curves on different synthetic topologies. In particular, a fitted midpoint can change while a coarse empirical threshold remains equal. Treating those summaries as interchangeable would misstate the comparison.

The lattice result makes the time horizon central to interpretation. Under repeated-attempt SI dynamics with positive transmission probability, connected networks can eventually spread far beyond the observed ten-round neighborhood. An apparent threshold within a short horizon need not be an eventual-spread threshold. Horizon sensitivity and size scaling must be studied directly before making broader claims about critical behavior.

The project is motivated by misinformation, but the simulated process does not know whether content is true, false, persuasive, or corrected. Its results apply to the specified mechanism on the specified graph models. Applying them to moderation policy or platform vulnerability would require data, parameter calibration, and validation outside this study.

5 Limitations and future work

The reference design uses five graph replicates, one network size, one time horizon, and a probability grid with step 0.05. These choices support an exploratory comparison rather than precise threshold inference. Confidence intervals are approximate and pointwise; there are no uncertainty estimates for fitted midpoints or empirical thresholds. Independent random sampling at each probability can introduce nonmonotonic fluctuations in finite-sample means.

The graphs are static and synthetic. Transmission probabilities are uniform, adoption is permanent, and the model omits recovery, user heterogeneity, recommender systems, repeated content competition, and content semantics. Approximate degree matching does not control for other structural differences. WS rewiring is not a pure clustering intervention.

Priority extensions are greater graph replication, finer grids around observed transitions, multiple horizons and sizes, explicit recovery or one-shot transmission variants, and degree-preserving structural comparisons. Real-data calibration and out-of-sample validation are prerequisites for predictive or platform-specific claims.

6 Reproducibility and availability

The implementation and all reference artifacts are available in the project repository. Python 3.12.14 and the dependency versions recorded in `results/metadata.json` produced the reference run. The reference command is:

```
misinformation-study --clustering --output results
```

The runner saves every trial, graph property, mean curve, fit status, random seed, and CSV SHA-256 hash. The notebook imports the same model rather than maintaining a separate implementation. Tests cover synchronous timing, quiet rounds, disconnected graphs, seeded replay, graph-level uncertainty, and end-to-end artifact generation.

This report's tables are generated from the checked-in CSVs after hash, trial-count, and parameter checks. `scripts/build_paper.sh` rebuilds the Markdown and PDF using `reports/paper.md.in`, Pandoc, and pdfLaTeX. Source templates and build instructions are included. Plot files are produced by the experiment runner. CSV replay under the recorded environment is deterministic; timestamps and rendered files need not be byte-identical.

7 Conclusion

Network topology changes mean finite-horizon adoption under the specified repeated-attempt SI process. The reference experiment shows earlier fitted growth in BA than ER, identical sampled 80% thresholds for those networks, and strong horizon constraints in low-rewiring WS networks. These observations support a reproducible modeling comparison without asserting universal misinformation dynamics or an isolated clustering effect.

References

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